Guide Before Commitment: When Does a Diffusion Model Decide What It Draws, and What Does Classifier-Free Guidance Change?

Author names and affiliations: *placeholder*

Pre-registered study; development split complete, test split partial

Abstract

Text-to-image diffusion models often produce plausible images that violate part of the prompt: an object is missing, a colour lands on the wrong object, or a count, a relative size or a spatial relation is wrong. We ask *when* along the denoising trajectory each kind of constraint stops depending on the remaining randomness, *what* classifier-free guidance (CFG) changes about that timing, whether cross-attention reveals it without labels, and whether a guidance schedule timed by the measurement is useful. On Stable Diffusion 1.5 and a controlled scene-graph benchmark we measure a chance-corrected restart-persistence curve $C(t)$ and its half-point t_{50} per constraint type by re-noising saved trajectory states. We also introduce Commitment-Aligned Guidance (CAG), which switches from high to low guidance after the development-set t_{50}. Four hypotheses, decision rules and all development-set constants were frozen in a signed protocol before any test image was generated. A blinded human audit validated the automatic judge only for object presence (accuracy 0.856); two development gates failed; we report all of this as found. Six of nine test shards exist, so every test result here is *partial and exploratory*. On them, stronger guidance makes object presence persistent earlier: t_{50} falls from 22.5 steps at $w = 2$ to 13.8 at $w = 15$ (registered contrast -8.67 steps, 95% CI $[-16.34, -4.72]$, Holm $p = 0.046$). The effect is not produced by the chance correction, but an independent-error stress test at the audited judge error removes it. Constraint-type orderings measured on the development split do not replicate on the test scenes, attention stabilisation does not track persistence (ordering concordance 0.30), and CAG is neither non-inferior to constant CFG nor better than its reversed schedule. Dose-matched controls explain CAG's diversity gain by its lower mean guidance rather than its timing. The protocol, estimator, judge audit and partial-test pipeline are documented so that the registered analysis can be completed unchanged once the missing shards exist.

1 Introduction

Prompt adherence of text-to-image diffusion models is usually summarised by one score per image. Compositional failures, however, are constraint-specific: an object can be present while its colour is wrong, or two objects can be correct while their left–right order is not. If different constraints are resolved at different moments of the sampling trajectory, three practical consequences follow. Guidance strength that matters early may be wasted late; diagnostics that read early internal signals are forecasts for some constraints and read-outs for others; and a pooled adherence score can hide both effects.

Prior work establishes that denoising stages control different image properties [1, 17, 23, 31], that guidance can be restricted or scheduled over time [16, 22, 27], and that cross-attention carries information about objects and attributes that can be read and steered [3, 5, 10, 29]. We therefore

do not claim that the general idea of stage-dependent generation is new. Our questions are narrower and measurement-first:

1. **When** does each constraint type (presence, count, colour binding, spatial relation, relative size) stop depending on the remaining randomness of the trajectory? We answer with an interventional, chance-corrected persistence curve rather than with an observational signal.
2. **What does CFG change?** Does a larger guidance scale move persistence earlier, once the higher seed-to-seed agreement it induces is accounted for?
3. **Can attention tell?** Does the step at which cross-attention stops changing track the interventional measurement?
4. **Is the timing usable?** Does a two-level guidance schedule whose switch step is read off development-set measurements keep adherence while raising diversity, and does it beat the same mean guidance applied in reverse order?

Contributions. (i) A two-probe measurement of restart persistence per constraint type with an explicit chance correction (Section 4); (ii) a controlled scene-graph benchmark with meaning-preserving and counterfactual variants, a factorial relative-size family and presence-conditional outcomes (Section 5); (iii) a blinded human audit of the automatic judge whose pre-declared criterion decides which constraint types can carry confirmatory claims (Section 6); (iv) Commitment-Aligned Guidance (CAG), a training-free schedule tested against constant, reversed, matched-mean and global-switch schedules (Section 4); (v) a pre-registered, signed protocol with development-set gates, all of which are reported whether they passed or failed (Section 8); and (vi) on the available test shards, robustness analyses that show which conclusions survive: a closed-form judge-error stress test of the persistence estimator, cross-split replication of commitment times, and dose-matched schedule contrasts (Section 10).

Findings in brief (partial test split, exploratory). Stronger guidance makes object presence persistent earlier, by about -8.67 of 40 steps between $w = 2$ and $w = 15$. This is the only registered rule that passed, and it is not an artefact of the chance correction. It does depend on the structure of judge errors: at the audited presence error rate, independent errors alone could produce it. The ordering of constraint types measured on the development split, which set CAG's switch steps, does not replicate on the test scenes, where colour binding becomes persistent after presence rather than before. Consistent with that mis-timing, CAG is not non-inferior to constant guidance, and its higher diversity is explained by its lower mean guidance: at equal mean guidance, spending guidance early is less diverse than spending it late. Attention stabilisation does not order constraint types the way persistence does. We use the operational term *restart persistence*; we do not claim that the model "decides" in any cognitive sense. Figure 1 summarises the probes and the schedule.

2 Related work

Diffusion sampling and guidance. We use denoising diffusion models [12], deterministic DDIM sampling [28], latent diffusion [26] and classifier-free guidance [11]. Restricting guidance to an interval of noise levels has been reported to improve sample quality [16], annealing the conditioning signal early to restore diversity [27], and choosing per-step guidance online with latent evaluators to be prompt-dependent [22]. Add-noise-then-denoise operations are used for editing and for sampling [19, 30]; we use re-noising only as a *measurement*.

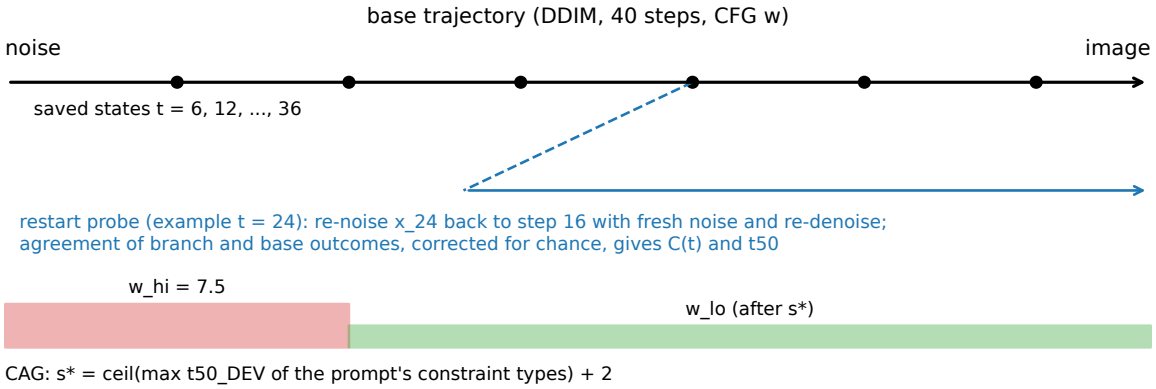


Figure 1: **Conceptual diagram** (not data). Top: a 40-step DDIM trajectory with saved states. Middle: the restart probe re-noises a saved state to an earlier step with fresh noise and re-denoises; agreement of branch and base outcomes, corrected for chance agreement between independent seeds, gives $C(t)$ and t_{50}. Bottom: CAG applies w_{hi} until a prompt-specific switch step $s^\star$ derived from development-set t_{50}, then w_{lo}.

Stages of generation. Switching prompts over denoising intervals localises layout, shape and detail to successive stages [23]; stage-specific text embeddings control different attributes [31]; an ensemble of stage-specialised denoisers is motivated by text conditioning mattering most early [1]; and features have been analysed theoretically as emerging in narrow time windows [17]. Strengthening the conditioning signal at selected steps has been reported to improve alignment more when done late than early [8], a finding in tension with front-loading guidance, and negative prompts to act with a delay [2].

Cross-attention. Cross-attention maps are used to interpret [29] and edit [10] generations, and attention-based objectives target object neglect and attribute binding [3, 6, 18, 25]; attention can also read and steer object size and position [5]. Initial-noise search and early prediction of image quality from attention are close in aim to our attention analysis [4, 9].

Evaluation. Object-focused and question-answering evaluators [7, 13–15] motivate our emphasis on judge reliability. Our judge combines open-vocabulary detection [20] with CLIP [24]; seed diversity uses DINOv2 embeddings [21].

Position of this work. Table 1 contrasts what the closest methods measure, intervene on and evaluate. The common thread is that timing matters; the differences are that we (a) measure timing *interventionally and per constraint type* with an explicit chance correction, (b) test whether CFG moves that timing, (c) compare an unsupervised attention signal against the interventional one, and (d) derive a guidance switch from the measurement rather than tuning it on adherence. To our knowledge these four elements have not been combined on controlled scene graphs; the literature search was narrow (Table 1 caption) and this is a candidate gap, not an established one.

Table 1: Closest prior work and this study. Entries describe each method at the level verified for this draft: full text for P2P and Attend-and-Excite; abstract or project page for the others (marked in `references.bib`). "–" means the aspect is not the method's aim.

Method	What it measures	What it intervenes on	How it is evaluated
Prompt-to-Prompt [10]	–	cross-attention maps during generation (injection, re-weighting)	image edits, qualitative and quantitative comparisons
Attend-and-Excite [3]	subject-token attention as a neglect signal	latents at inference, to raise neglected tokens' attention	object presence/binding via CLIP-based metrics and a user study
Prompt Mixing [23]	–	prompt used in different denoising intervals	localisation of shape variations
Critical windows [17]	theoretical time windows of feature emergence	–	theory and experiments
Interval guidance [16]	–	CFG applied only in a limited noise interval	distribution and sample quality
CADS [27]	–	annealed (noised) conditioning during sampling	diversity and quality
Dynamic CFG [22]	per-step feedback from latent evaluators	per-step guidance scale, chosen online	prompt adherence and quality
Step selection [8]	–	conditioning signal strengthened at selected denoising steps (later steps favoured)	text–image alignment
This study	chance-corrected restart persistence $C(t)$, t_{50} per constraint type; attention stabilisation t_{stab}	two-level CFG schedule switched at $s^\star$ derived from DEV t_{50} (CAG)	pre-registered tests with a human-audited detector judge; constant, reversed, matched-mean and global-switch controls

3 Hypotheses and pre-registration

The four hypotheses form one Holm family ($\alpha = 0.05$). A hypothesis is reported *supported* only if its own decision rule passes *and* its Holm-adjusted $p < 0.05$, with Holm computed over all four hypotheses even when one is not estimable. Definitions, decision rules and development-set constants were fixed in Addendum C and a signed protocol file before any test image was generated (103 code, configuration, benchmark and documentation files hashed; Section 15).

HA, constraint-specific persistence. Primary estimand $\Delta t_{50} = t_{50}^{\text{bind|pres}} - t_{50}^{\text{pres}}$ at $w = 7.5$ with the fixed-jump restart (D8). Support: the 95% scene-bootstrap interval excludes 0, $|\Delta t_{50}| \geq 3$ steps, and the joint-binding companion and the equal-ratio restart (ER30) agree in sign. Bounded null: the interval lies inside $(-1.5, 1.5)$.

HD, CFG moves persistence. $\Delta t_{50}^{\text{CFG}} = t_{50}(w{=}15) - t_{50}(w{=}2)$ for presence and for binding given presence, with chance agreement computed separately at each w and one shared scene re-

Table 2: Status of each hypothesis fixed before the test split (Addendum C, `configs/analysis_plan_r3.json pre_test_status`); applied automatically by the analysis code.

Hyp.	Registered rule	Status for the test split
HA	Δt_{50} rule + Holm	EXPLORATORY: DEV precision gate failed (half-width 7.94 $>$ 4); binding judge below criterion
HD	$\Delta t_{50}^{\mathrm{CFG}}$ rule + Holm	confirmatory through presence only; binding estimand exploratory
HE	intersection–union + Holm	registered rule unchanged; F includes judge types below the criterion (disclosed); presence-only F as exploratory sensitivity
HF	concordance/Spearman rule + Holm	registered rule unchanged; type orderings include judge types below the criterion (disclosed)

sample. Supported if at least one estimand is negative with $|\Delta| \geq 3$ and an interval excluding 0; $p_{\mathrm{HD}} = \min(1, 2 \min p)$. A positive interval is reported as contradicting HD.

HF, attention stabilisation. Supported if the ordering of constraint types by t_{stab} is concordant with their ordering by t_{50} (concordance ≥ 0.8) and the within-type Spearman correlation between scene t_{stab} and the scene's area under $C(t)$ has a 95% interval above 0 for at least two types.

HE, Commitment-Aligned Guidance. Supported only if all three hold (intersection–union, p = maximum of the three one-sided p): adherence F of CAG (A2) is non-inferior to constant CFG 7.5 (A1) (90% interval lower bound > -0.03), seed diversity is higher (95% interval > 0), and A2 beats the reversed schedule A4 with equal mean guidance on F (95% interval > 0).

Status fixed before the test split. Development-set evidence (Section 8) fixed the following before any test image existed (Table 2): HA is exploratory because its precision gate failed; per-type results for count, binding, spatial relation and relative size are exploratory because their judge accuracy missed the audit criterion; HD can be supported only through its presence estimand; HE and HF keep their registered rules, and their reports disclose that F and the type orderings include judge types below the criterion.

4 Methods

Generator. Stable Diffusion 1.5 [26] at a pinned revision in fp16, DDIM [28] with $\eta = 0$ and $T = 40$ steps at 512^2 pixels, empty negative prompt. With text condition c and guidance scale w, the noise prediction at step i is the classifier-free combination [11]

$$\hat{\varepsilon}_i = \varepsilon_\theta(x_i, \varnothing) + w_i \left(\varepsilon_\theta(x_i, c) - \varepsilon_\theta(x_i, \varnothing)\right), \tag{1}$$

where w_i is constant ($w_i = w$) except under the guidance schedules below. When $w_i = 1$ only the conditional branch is evaluated, and the number of network evaluations is counted per image.

Outcomes. An automatic judge (Section 6) scores each image for each constraint κ of its prompt: presence (all required objects detected), count (exact), and, conditional on the relevant objects being present, colour binding, left/right relation and relative size (box area of the intended larger object at least 1.5× the other's). Conditional outcomes are undefined, not zero, when an object is missing. The adherence score F of an image is the mean of the outcome families present.

Restart probe. The latent after $t \in \mathcal{G} = \{6, 12, 18, 24, 30, 36\}$ completed steps is saved. A branch re-noises it to an earlier step index j with fresh noise $\varepsilon \sim \mathcal{N}(0, I)$ and denoises deterministically to the end at the base's guidance:

$$x'_j = \sqrt{r}\, x_t + \sqrt{1-r}\, \varepsilon, \qquad r = \bar{\alpha}(j)/\bar{\alpha}(t), \tag{2}$$

with $\bar{\alpha}(s)$ taken at the scheduler timestep of step s. The fixed-jump probe (D8) uses $j = \max(t - 8, 0)$, whose noise share $1 - r$ grows with t; the equal-ratio probe (ER30) chooses j so that $r \approx 0.30$ at every t; a full resample (FULL) restarts from pure noise and serves as a validity control.

Chance-corrected persistence. For scene s and constraint type κ, let $A_{\kappa s}(t)$ be the share of (base, branch) pairs with equal outcomes and $A0_{\kappa s}$ the agreement between independent seeds of the *same base prompt, guidance and step count*. Persistence pools scenes as a ratio estimator,

$$C_\kappa(t) = \frac{\sum_s \left[A_{\kappa s}(t) - A0_{\kappa s}\right]}{\sum_s \left[1 - A0_{\kappa s}\right]}, \tag{3}$$

so $C = 0$ means no more agreement than an independent sample and $C = 1$ means the outcome is fixed. t_{50}^κ is the linearly interpolated first crossing of 0.5; it is left-censored when C already exceeds 0.5 at $t = 6$ and right-censored when it never does. Because the noise share of D8 varies with t, absolute t_{50} is relative to the probe; ER30 is the registered sensitivity check.

Prompt-switch probe. A counterfactual trajectory (colour swap, relation reversal or size swap) is continued from its saved state with the canonical prompt, and

$$C^{\text{switch}}(t) = \frac{\sum_s \left[W_s(t) - W0_s\right]}{\sum_s \left[W1_s - W0_s\right]}, \tag{4}$$

where W is the share of switched images satisfying the counterfactual constraint and $W0$, $W1$ are that share among canonical and counterfactual images.

Attention stabilisation. Cross-attention maps of the constraint words are recorded at every step (conditional branch, averaged over heads and over layers within the 16×16 and 32×32 groups; recording leaves the image unchanged). Per type a signal a_i is formed (noun map, colour-word map, ratio of effective areas, angle between noun centroids, connected components), its velocity $v_i = 1 - \cos(a_i, a_{i-1})$ (or the absolute change of a scalar), and t_{stab} is the step at which half of the cumulative path $\sum_i v_i$ has been travelled.

Commitment-Aligned Guidance. For a prompt with constraint types K, the switch step and the schedules are

$$s^\star = \min\Big(\big\lceil \max_{\kappa \in K} t_{50,\text{DEV}}^\kappa \big\rceil + 2,\ T\Big), \qquad w_i^{\text{A2}} = \begin{cases} w_{hi} & i < s^\star \\ w_{lo} & i \geq s^\star \end{cases}, \qquad w_i^{\text{A4}} = \begin{cases} w_{lo} & i < s^\star \\ w_{lo} + \frac{s^\star (w_{hi} - w_{lo})}{T - s^\star} & i \geq s^\star \end{cases}, \tag{5}$$

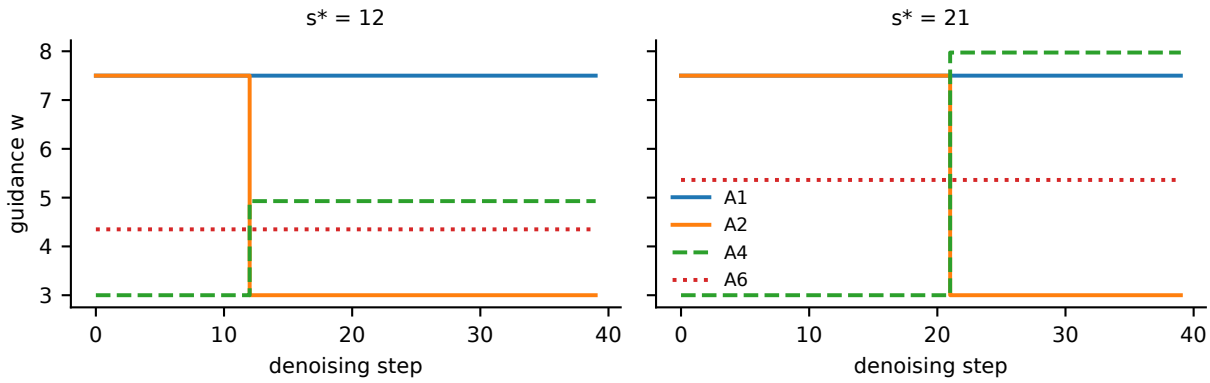


Figure 2: Exact guidance schedules of the frozen design (computed by `src/cag.py`) for $s^\star = 12$ and $s^\star = 21$ with $w_{lo} = 3$: constant CFG (A1), CAG (A2), reversed with equal mean (A4) and constant at A2's mean (A6). A4 is not capped; its maximum guidance is recorded per job.

with $w_{hi} = 7.5$. A4 has the same mean guidance as A2 in reverse order; A6 applies that mean as a constant; A3 uses a global $s^\star$ from the median DEV t_{50}; A1 is constant 7.5. The frozen design uses $w_{lo} = 3$, selected by the DEV quality guard (Section 8); Figure 2 shows the exact schedules for the two switch steps that occur on the test split.

Statistical analysis. Uncertainty uses a scene-level bootstrap with 10,000 draws and fixed random keys, resampling scenes (not images) so that seeds and branches of a scene stay together; p-values are bootstrap tail probabilities. Holm's procedure controls the family-wise error over HA, HD, HE and HF, with a non-estimable hypothesis counting as $p = 1$. Identical generations (same prompt text, guidance, steps and seed in one split) are counted once, two canonical scenes with identical text count once, and test prompts whose text also occurs in the development split are excluded from variant-level summaries.

Sensitivity to judge error (exploratory, not registered). Restart persistence is a statement about agreement between judged outcomes, so judge errors enter both A and $A0$. If each judged outcome is flipped independently with probability e (symmetric, non-differential error), every observed agreement is the affine image $A^{\text{obs}} = (1-2e)^2 A + c$ of the true agreement with $c = 2e(1-e)$, for branch pairs and seed pairs alike. Substituting into the pooled estimator gives the error-corrected persistence

$$C_\kappa^{(e)}(t) = \frac{\sum_s \left[A_{\kappa s}^{\text{obs}}(t) - A0_{\kappa s}^{\text{obs}}\right]}{\sum_s \left[1 - c - A0_{\kappa s}^{\text{obs}}\right]}, \tag{6}$$

which equals the registered $C_\kappa(t)$ at $e = 0$. Error makes C shrink towards zero by an amount that depends on $A0$, which differs between guidance scales. The model assumes errors independent across images; a detector's errors are usually content-dependent (the same hard object is missed in every seed), which can move agreement in either direction and is not captured. We therefore use Eq. (6) as a stress test of how much judge noise a conclusion tolerates, not as a correction of the estimate, and evaluate $e \in \{0.05, 0.10\}$ and $e = 1 -$ (audited presence accuracy).

Table 3: Benchmark V2 design counts (scene graphs are the independent unit). TEST counts are design metadata only; no TEST image exists.

Split	Scenes	Binding-bearing (literal)	Competitive subset	Spatial-bearing	Prompt variants
DEV	100	70	55	28	470
TEST	200	140	110	54	940

5 Benchmark

Prompts are generated from machine-readable scene graphs with a grammar and a lexicon. Every scene has a canonical prompt (CAN), meaning-preserving variants (synonym paraphrase PAR, mention-order change ORD) and counterfactual variants that change exactly one declared field (colour swap CF-C, relation reversal CF-R, count change CF-N, object deletion CF-O, attribute insertion or removal CF-A). Scenes are split into DEV and TEST at scene level, with object-pair groups confined to one split. The main benchmark has 100 DEV and 200 TEST scenes and 1,410 prompts (some variant types do not apply to some scene types). A factorial family of 24 object pairs (8 DEV, 16 TEST) crosses colour, relative size and spatial relation on top of presence, in the style of the task's example "a small red cube and a large blue sphere" (192 canonical prompts plus colour-swap, size-swap and relation-reversal counterfactuals).

Variant review and amendment. All 1,674 prompts and 73 lexicon entries were reviewed against the scene-graph constraints before generation. Two synonym choices were replaced in a dated amendment before any image of the affected prompts was generated: "golden" for yellow (the judge's text encoder placed it nearest to brown) and "on the left/right side of" (which admits the reference object's own frame); identifiers, constraints and splits were unchanged.

6 Evaluation and judge audit

The judge (J1) combines open-vocabulary detection [20] with CLIP colour classification [24] at pinned model revisions, with thresholds fixed before any test data (presence 0.10, count 0.15, size ratio 1.5). A second, question-answering judge was planned but not implemented and plays no role in any result.

Pre-declared audit. A blinded, seeded, stratified sample of DEV (image, constraint) items was labelled by a human who saw only an opaque image and a neutral question (a colour, a count, a presence answer, which object is further left, or which looks larger), with "not visible" and "ambiguous" answers. Accuracy is population-weighted with a stratified bootstrap. A constraint type may carry a confirmatory claim only if its weighted accuracy is at least 0.85; no other criterion was added after the data were seen.

Result. Only presence met the criterion (Table 4, Figure 3). The binding judge reached 0.800, count 0.768, spatial relation 0.615 (its sensitivity was low) and relative size 0.846. One annotator labelled the main package; a second package was labelled by the same person within the same session and is therefore neither an inter-rater nor a valid intra-rater check, and is not used. These limits are carried into every result (Section 3).

Table 4: DEV evaluator audit of J1 against blinded human labels (one annotator; population-weighted accuracy, stratified bootstrap 95% CI). Only types meeting ≥ 0.85 may carry a confirmatory claim.

Type	n	Accuracy	95% CI	≥ 0.85
presence	159	0.856	[0.802, 0.907]	yes
count	108	0.768	[0.685, 0.843]	no
binding	156	0.800	[0.722, 0.871]	no
spatial	109	0.615	[0.523, 0.706]	no
size	175	0.846	[0.794, 0.897]	no

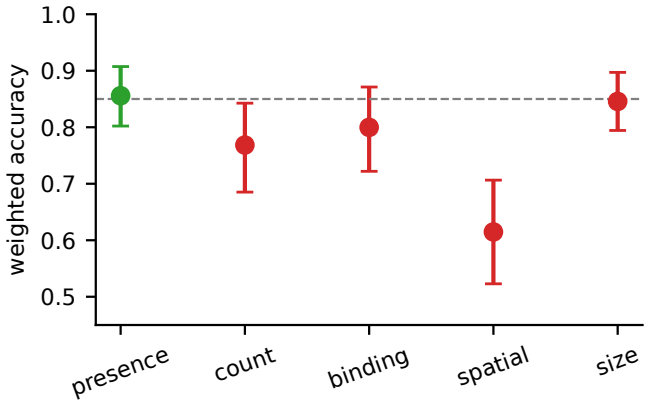


Figure 3: Weighted accuracy of the automatic judge per constraint type on DEV (95% bootstrap intervals; one annotator). Dashed line: pre-declared criterion 0.85. Green: met; red: not met.

7 Experimental design and execution

All jobs come from deterministic manifests sharded by scene group over three workers (Table 5). A stage guard refuses any stage whose owner-controlled gates are false, and test jobs are refused unless the signed protocol verifies against the code, configurations, benchmark, judge definition and development-set constants. Every worker output is merged against its manifest before analysis; missing, duplicated, corrupt or mismatching outputs are rejected, never silently dropped. The development stages ran first: the canonical DEV core (400 images: 100 prompts, four seeds, saved states; restarts branch from seeds 0 and 1), the DEV variants and size family (1,784 jobs), restart branches and CFG bases (8,896 jobs) and the CAG quality guard (400 jobs). The test split comprises 9,160 core and CFG×steps jobs, 10,560 restart and switch jobs, and 5,536 CAG and size-family jobs (19,569 full-generation equivalents). Generation used two NVIDIA T4 GPUs at about 7.0 s per 40-step image per GPU.

8 Development-split results

Everything in this section uses DEV scenes only and was fixed before any test image was generated. It is exploratory; its role is to set constants and to decide, through pre-declared gates, which claims the test split can carry.

Table 5: Frozen stage manifests (job counts and full-generation equivalents from `src/manifests.py` with the frozen constants).

Split	Stage	Content	Jobs	Equiv.
DEV	DEV_CORE	variants, CF bases, DEV size family	1,784	1,784
DEV	DEV_COMMIT	restarts D8/ER30/FULL, CFG 2/15 bases and restarts, switches	8,896	6,930
DEV	DEV_CAG_GUARD	CAG $w_{lo} \in \{1, 3\}$ quality guard	400	336
TEST	TEST_CORE	all variants; CFG × steps	9,160	7,660
TEST	TEST_COMMIT	restarts at CFG 2/7.5/15; prompt switches	10,560	6,986
TEST	TEST_CAG_SIZE	CAG arms A2/A3/A4/A6; size family with restarts and switches	5,536	4,923

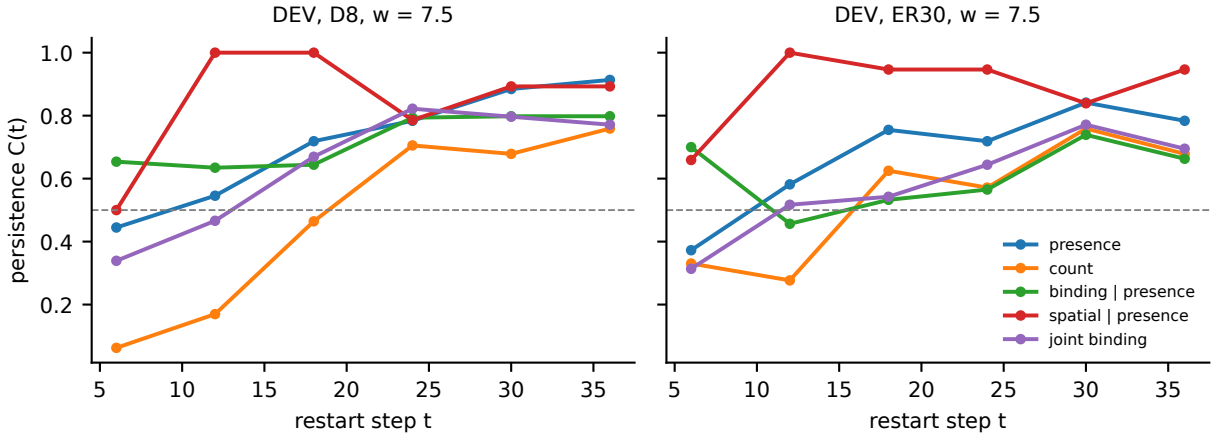


Figure 4: DEV restart persistence $C(t)$ at $w = 7.5$ for the fixed-jump (D8) and equal-ratio (ER30) probes (corrected estimator; exploratory). Dashed line: $C = 0.5$.

An estimator error found and corrected before the test split. The chance-agreement term $A0$ had first been computed from every core image of a scene, including paraphrases, counterfactual variants and other guidance scales, instead of only independent seeds of the branches' own base prompt as specified. This lowers $A0$ and biases every $C(t)$ upwards. The code was corrected, a regression test added, and the DEV constants were recomputed from the stored judgements; both versions are archived. All DEV numbers below use the corrected estimator.

Persistence curves. Figure 4 shows $C(t)$ on 100 DEV scenes. Presence crosses 0.5 near $t_{50} = 9.29$ steps (D8); colour binding given presence is already above 0.5 at the first grid step (left-censored, $t_{50} \leq 6.0$), and count is latest (18.89). With both restart probes the binding-minus-presence difference is negative but its interval includes zero: D8 $\Delta t_{50} = -3.29$ $[-7.56, 8.32]$, ER30 -3.66 $[-6.83, 6.77]$ (Table 6). Left-censoring means DEV cannot place binding relative to presence; and the binding judge missed the audit criterion.

Gates. Table 7 lists every pre-declared DEV gate. The HA precision gate failed (half-width 7.94 steps against a limit of 4), so HA is exploratory on the test split by the pre-declared rule. The full-resample control failed (maximum $|C| = 0.568$ against 0.15): the statistic is a maximum over

Table 6: DEV persistence steps t_{50} at $w = 7.5$ (EXPLORATORY; 163 scenes). †left-censored: already above 0.5 at the first grid step.

Kind	D8	ER30
presence	9.3	9.7
count	18.9	15.8
binding | presence	6.0†	6.0†
spatial | presence	6.0	6.0†
joint binding	13.0	11.5
Δt_{50} (binding − presence)	-3.29 [-7.56, 8.32]	-3.66 [-6.83, 6.77]

Table 7: DEV gates (corrected estimator). Failed gates are reported as failed; HA is therefore exploratory.

Gate	Criterion	DEV value	Result
HA precision	CI half-width ≤ 4 steps	7.94	failed
Full-resample control	$\max \lvert C \rvert \leq 0.15$	0.57	failed
HD presence, $w = 2$	≥ 40 scenes	45	passed
HD presence, $w = 15$	≥ 40 scenes	59	passed
CAG quality (HPS v2), $w_{lo} = 1$	mean drop ≤ 0.5	-0.52	failed
CAG quality (HPS v2), $w_{lo} = 3$	mean drop ≤ 0.5	-0.26	passed

30 cells, several resting on fewer than ten scenes, so even exactly independent outcomes rarely meet it at this sample size; but presence, the best-supported cell, shows a consistent offset of about +0.16 that sampling error does not explain and for which no cause was found in the code. We report it as an open validity concern for the restart estimator. Both HD presence gates passed (45 and 59 scenes).

CAG quality guard. HPS v2 scored the 200 DEV guard images per w_{lo} against their constant-CFG partners (same prompt and seed). With $w_{lo} = 1$ the mean change was -0.518 HPS points (95% interval -0.622 to -0.418), a drop just beyond the pre-declared 0.5; with $w_{lo} = 3$ it was -0.259 (-0.324 to -0.204). By the pre-declared rule $w_{lo} = 3$ was adopted before the test split (Figure 5). The switch steps derived from DEV are $s^\star = 12$ for prompts whose latest type is presence, binding or spatial relation and $s^\star = 21$ for prompts with counts (global $s^\star = 12$).

9 Partial TEST results (exploratory)

Generated by `partial/partial_report.py` from the verified TEST shards that exist (archive r3-test-partial-20261020T134300Z-40ade4d; judged with torch 2.11.0+cu128 on 2 GPU(s)). Every quantity below is **exploratory**: it covers a subset of the TEST scenes and is not the registered confirmatory test.

Limitations of this analysis.

- Missing TEST shards (not imputed): TEST_CORE shard 2; TEST_COMMIT shard 2; TEST_CAG_SIZE shard 0; estimates cover only the scene groups of the present shards.
- This is not the registered confirmatory analysis: that requires all nine shards and notebook 04; Holm-adjusted p-values are shown for completeness only.

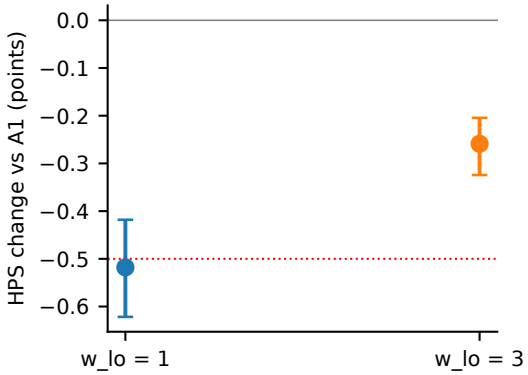


Figure 5: DEV CAG quality guard: mean HPS v2 change of CAG images relative to constant CFG 7.5 (95% scene-bootstrap intervals). Dotted line: pre-declared limit −0.5.

Table 8: TEST coverage of this PARTIAL TEST, EXPLORATORY analysis. Shards are disjoint scene groups; missing shards are not imputed.

Stage	Shards present	Missing	Jobs merged	Scenes
TEST_CORE	[0, 1]	[2]	6172/6172	133
TEST_COMMIT	[0, 1]	[2]	7080/7080	85
TEST_CAG_SIZE	[1, 2]	[0]	3750/3750	222

- HA is exploratory by its failed DEV precision gate; the full-resample control failed on DEV (Section on DEV results).
- Only the presence judge met the audit criterion (single annotator, no valid double annotation); count, binding, spatial and size outcomes are exploratory.
- HE uses only scenes with both A1 and CAG images (67 scenes).

10 Partial TEST: robustness, replication and interpretation

The analyses in this section were *not* registered. They were chosen after the registered rule outcomes of Section 9 were known, to probe the threats a sceptical reader would raise first, and they are computed by `partial/robustness.py` from the saved judged rows of the same archive with the frozen estimator code; no image was generated or re-judged. All quantities are partial-TEST and exploratory.

Evidence integrity. The archive merges exactly the jobs of the present shards' manifests (Table 8); every merged image was judged once and has one DINOv2 embedding (17,002 each, unique identifiers), and the frozen de-duplication rules then remove 28 repeated generations. A second, independent judging of the same images on different hardware and software (one consumer GPU with PyTorch 2.6 instead of two data-centre GPUs with PyTorch 2.11) reproduced every judged outcome on all 16,974 de-duplicated rows (100.00%), and all four registered rule outcomes; bootstrap p-values differed by at most 0.002. Two numerical warnings raised during the analysis come from scenes without an eligible pair for a constraint type and from degenerate bootstrap resamples in

Table 9: Hypotheses on the available TEST shards (PARTIAL TEST, EXPLORATORY). p_{Holm} over the registered family is shown for completeness only; no result here is a confirmatory finding.

Hyp.	Estimand	Estimate	CI	p	p_{Holm}	Rule outcome on partial data
HA	Δt_{50} binding−presence (steps)	3.59	[-10.43, 13.06]	0.5652	1.0000	EXPLORATORY (DEV precision gate failed (half-width 7.94 > 4); binding judge below the audit criterion); rule outcome: inconclusive
HD	Δt_{50} presence, $w15{-}2$ (steps)	-8.67	[-16.34, -4.72]	0.0116	0.0464	supported
HE	A2−A1 on F (90% CI; margin −0.03)	-0.016	[-0.035, 0.003]	0.4825	1.0000	not supported (non_inferior, beats_reversed failed)
HF	type-order concordance	0.30	[0.20, 0.75]	0.8042	1.0000	rejected (ordering concordance ≤ 0.5)

Table 10: Persistence steps t_{50} at $w = 7.5$ on TEST shards present (PARTIAL TEST, EXPLORATORY; HA is exploratory by its DEV gate). †left-censored.

Kind	D8	ER30
presence	16.2	13.3
count	15.3	14.0
binding | presence	19.8	21.0
spatial | presence	6.0†	6.0†
size | presence	13.6	–
joint binding	21.2	21.3
Δt_{50} (binding−presence) [95% CI]	3.59 [-10.43, 13.06]	7.67 [-12.64, 24.88]
scenes	111	67

the HF Spearman statistic; the frozen code treats both as missing, which leaves the point estimates unchanged but makes the relative-size Spearman interval ($[-1, 1]$) uninformative.

What CFG changes about timing (HD). Presence persistence moves monotonically earlier with guidance: $t_{50} = 22.5$ steps at $w = 2$, 16.2 at $w = 7.5$ and 13.8 at $w = 15$ (Table 16, Figure 10a). Two checks argue against the effect being produced by the chance correction itself. Chance agreement between independent seeds is not higher at $w = 15$ ($\bar{A}_0 = 0.697$) than at $w = 2$ (0.729), so the correction does not favour high guidance; and the uncorrected branch–base agreement at $t = 18$ is already higher at $w = 15$ (0.90) than at $w = 2$ (0.79). The registered bootstrap resamples all 111 scenes of the guidance levels, of which only 67 have branches at both $w = 2$ and $w = 15$; resampling only those gives the same estimate with interval $[-16.01, -4.52]$ (Table 17). The sign agrees in both TEST_CORE shards and in four of five scene categories; in the 10 counting scenes the contrast reverses (+11.4 steps), so the effect is not uniform across prompt types.

The conclusion is fragile in one specific way. Under the independent-error stress test of Eq. (6) the estimate stays negative, but its interval reaches zero already at $e = 0.05$ (-12.91, $[-15.49, 0.33]$) and at $e = 0.10$; at the audited presence error rate ($e = 0.144$) every corrected curve exceeds 0.5 at the first grid step, so t_{50} is left-censored at all guidance scales and the contrast is not identifiable (Figure 10b). The HD result therefore holds only if the presence judge's errors are

Table 11: HD on the available shards (PARTIAL TEST, EXPLORATORY): t_{50} by guidance scale; binding estimand exploratory by the audit.

Estimand	t_{50} (w=2 / 7.5 / 15)	Δt_{50} (15−2)	95% CI	p	rule
presence	22.5 / 16.2 / 13.8	-8.67	[-16.34, -4.72]	0.0058	supported
binding | presence	>36 / 19.8 / 12.5	-inf	[-24.00, 1.50]	0.0742	inconclusive

Table 12: HE: CAG ($w_{lo} = 3$) vs constant CFG 7.5 (A1) and the reversed schedule A4, on scenes with both A1 and CAG images (PARTIAL TEST, EXPLORATORY; 67 scenes).

Contrast	Estimate	CI	Requirement	Met
A2−A1 on F	-0.016	[-0.035, 0.003]	90% CI lower > −0.03	no
A2−A1 diversity	0.0219	[0.0151, 0.0286]	95% CI > 0	yes
A2−A4 on F	0.001	[-0.025, 0.027]	95% CI > 0	no
A3−A1 on F	-0.031	[-0.056, -0.006]	secondary	–
A6−A1 on F	-0.005	[-0.041, 0.029]	secondary	–

mostly content-dependent rather than independent across seeds. That is plausible for a detector but is not established by our audit, which measured accuracy, not the error structure.

What CFG changes about adherence. On the 102 scenes generated at every guidance level and step count, the presence rate rises from 0.32 at $w = 2$ to 0.43 at $w = 15$ and the rate of images satisfying every constraint from 0.09 to 0.17, with most of the gain reached by $w \approx 5$–10; halving the number of steps to 20 changes neither curve beyond its uncertainty (Figure 11). Even at the best guidance scale fewer than one image in five satisfies its whole prompt, which is the failure rate the paper set out to explain.

Commitment times do not replicate from DEV to TEST. Table 18 compares t_{50} at $w = 7.5$ on the two splits. DEV placed colour binding at or before the first grid step, presence at 9.3 and count last (18.9); the partial TEST places count (15.3) and presence (16.2) before binding (19.8), and joint binding moves from 13.0 to 21.2. The equal-ratio probe shows the same shift, so it is not specific to the D8 noise schedule. The DEV presence value lies outside the TEST bootstrap interval. We cannot attribute the shift: the TEST estimate covers two of three shards, DEV failed its own precision gate, and the binding judge is not validated. The shift does, however, matter for CAG: its switch steps were read off DEV ($s^\star = 12$ for presence, binding and spatial prompts), and on the TEST scenes binding is not yet persistent at step 12, so A2 lowers guidance before the constraint it is meant to protect is settled.

CAG: dose, not timing (HE). The registered HE contrast compares CAG with constant $w = 7.5$, but every CAG arm applies less guidance on average ($\bar{w} = 4.64$ for A2, A4 and A6). The dose-matched controls separate the two factors (Table 19, Figure 12). The diversity gain of A2 over A1 (+0.022) is matched by a constant schedule at the same mean (A6, +0.027) and exceeded by the reversed schedule (A4, +0.040); at equal mean guidance, A2 is *less* diverse than A6 (-0.013, $[-0.026, -0.001]$) and A4 (-0.030), and its adherence does not differ from A6 ($\Delta F = -0.002$, $[-0.026, 0.023]$). The one-switch-for-all variant A3 lowers adherence ($\Delta F = -0.031$, $[-0.057, -0.008]$). The pre-declared presence-only sensitivity of A2 against A1 is -0.026 ($[-0.052, 0.000]$). On these data, concentrating guidance before the DEV commitment step buys

Table 13: Mean UNet evaluations per image by arm (PARTIAL TEST, EXPLORATORY).

Arm	UNet calls
A1	80.0
A2	80.0
A3	80.0
A4	80.0
A6	80.0

Table 14: HF: attention stabilisation vs persistence (PARTIAL TEST, EXPLORATORY).

Quantity	Value
concordance	0.30
95% CI	[0.20, 0.75]
pairs	10
Spearman ρ, binding | presence	0.20 [-0.37, 0.69]
Spearman ρ, count	-0.30 [-0.83, 0.43]
Spearman ρ, presence	-0.19 [-0.42, 0.07]
Spearman ρ, size | presence	0.12 [-1.00, 1.00]
Spearman ρ, spatial | presence	0.37 [-0.43, 0.89]
rule	rejected (ordering concordance ≤ 0.5)

nothing that the same amount of constant guidance does not, and costs diversity relative to spending it late, in line with reports that guidance at high noise levels reduces diversity [16] and that late conditioning helps alignment [8].

11 Registered confirmatory analysis

PENDING (the registered confirmatory analysis requires all nine test shards; TEST_CORE shard 2, TEST_COMMIT shard 2 and TEST_CAG_SIZE shard 0 are missing). When they exist, `scripts/final_report.py` verifies the archive and the signed protocol, refuses any non-confirmatory input, and writes `sections/results_test.tex` with `tables/test_hypotheses.tex` (fields `results.{HA,HD,HE,HF}.{p, p_holm, final}`), `test_ha/hd/he/hf/s2.tex` and `figures/test_*.pdf`. No confirmatory claim is made in this paper before that.

12 Earlier evidence: pilot and infrastructure (exploratory)

This section reports the pilot that motivated the study, (A), the infrastructure and integrity checks of the first cloud runs, (B), and a first-pass look at the DEV core made before the judge audit, (C). None of it tests the registered hypotheses.

12.1 A. Pilot (exploratory)

The pilot generated 1008 images (28 prompts, three CFG scales, two step counts, 6 seeds). Table 20 and Figure 13 show constraint satisfaction; CFG contrasts are in Table 21. With only 4–6 prompts per category the intervals are wide and no CFG effect on adherence is resolved; absence of a resolved effect is not evidence of its absence. Global attention re-weighting (scale 2 against the control path) changed overall adherence by +0.02 (95% CI -0.04 to +0.09), i.e. no resolved

Table 15: S2 (EXPLORATORY): outcome rates by prompt variant at $w = 7.5$ on the available TEST shards; 28 rows whose prompt occurs in DEV excluded.

Variant	rows	objects_present	count_ok	bind_given_pres	spatial_given_pres	size_given_pres
CAN	884	0.391	0.257	0.434	0.551	0.494
CF-A	320	0.428	0.362	0.726	–	–
CF-C	314	0.255	0.235	0.237	0.652	0.500
CF-N	148	0.662	0.297	0.622	–	–
CF-O	104	0.981	–	–	–	–
CF-R	166	0.355	–	0.269	0.390	0.222
CF-S	22	0.500	–	0.273	0.818	0.364
ORD	452	0.327	0.132	0.271	0.479	–
PAR	452	0.343	0.277	0.450	0.605	–

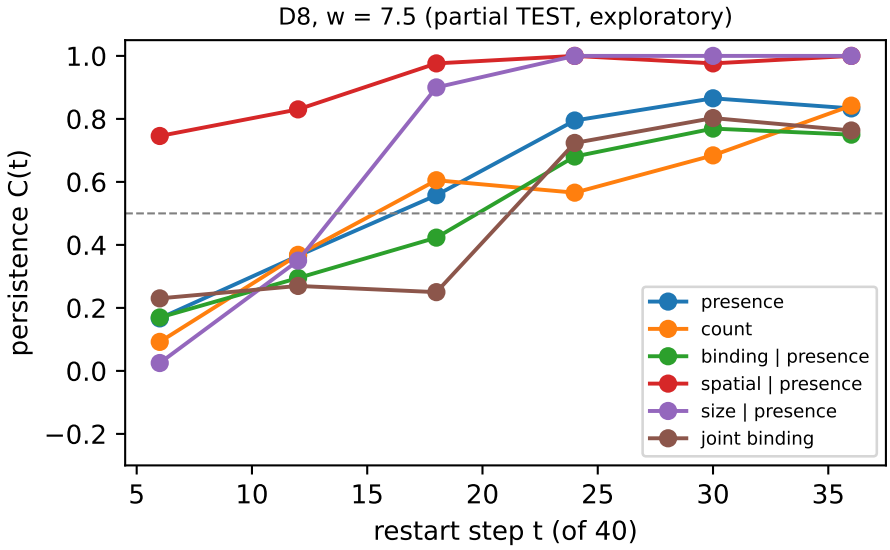


Figure 6: Chance-corrected persistence $C(t)$ by prompt type under the D8 restart probe ($w = 7.5$); dashed line marks $C = 0.5$, whose crossing defines t_{50}. Partial TEST, exploratory.

improvement (Table 22). Among binding-bearing pilot images, all required objects were detected in 395 of 648 (0.61), but only 179 of 432 (0.41) for the binding and multi categories, because counting prompts are always eligible; the pilot's legacy binding score already counts omissions as failures. In prompt-grouped cross-validation, early attention features added ΔAUC of +0.050 (95% CI +0.017 to +0.097) over the design and seed baseline, whereas seed identity added +0.002 (Table 23); only six distinct noise vectors exist in the pilot, a seed-0 deficit appeared (permutation $p = 0.012$), and both are post hoc observations that motivate HB without testing it.

12.2 B. Infrastructure and integrity (not scientific findings)

On a Tesla T4 with the committed code, the GPU identity script ran 3 cases and all six required identities held (determinism, loop from step 0, reload-and-resume, zero-magnitude branch, hook-off control, restored baseline; maximum pixel difference 0). The scope is three pilot prompts at one seed and setting; the model revision is not recorded inside that artifact. The R2a DEV core contains

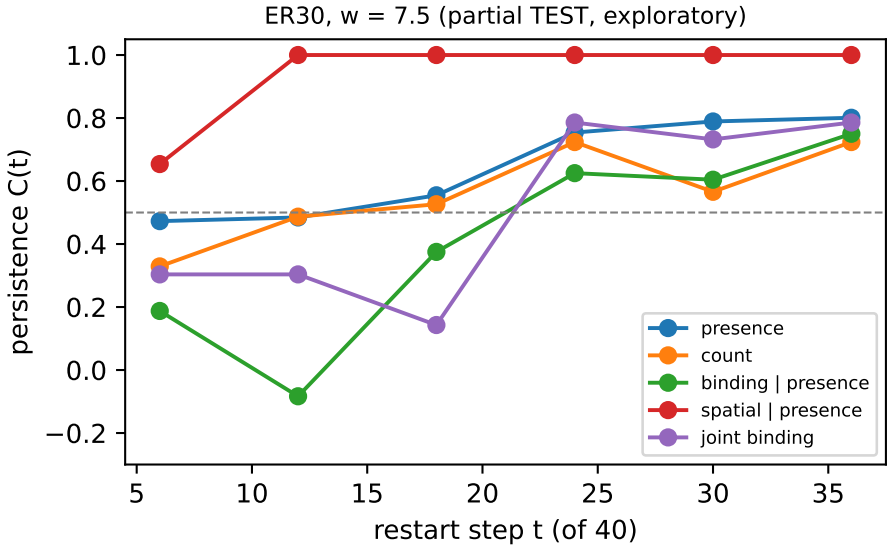


Figure 7: Chance-corrected persistence $C(t)$ by prompt type under the ER30 restart probe ($w = 7.5$); dashed line marks $C = 0.5$. Partial TEST, exploratory.

Table 16: Commitment time by guidance scale on canonical prompts (restart D8, 40 steps; partial TEST, exploratory). $\bar{A}_0$: mean chance agreement of independent seeds.

w	t_{50} presence	95% CI	$\bar{A}_0$ presence	t_{50} binding|pres.
2	22.5	[19.4, 28.0]	0.729	–
7.5	16.2	[12.4, 19.1]	0.649	19.8
15	13.8	[7.1, 17.2]	0.697	12.5

"–" marks a curve that never reaches $C = 0.5$ on the grid (right-censored, $t_{50} > 36$).

400 images and 2400 saved checkpoints; an independent audit passed 22 of 22 hard checks (manifest coverage, hashes, readability, no duplicates or orphans, checkpoint layout). A one-sample-pair regeneration on a different GPU matched the stored latent at the same step best, supporting the checkpoint indexing; this was not a bit-level Kaggle comparison. Two throughput probes disagree by about 25% (Table 24) and are not reconciled.

12.3 C. First-pass DEV outcomes (unaudited judge)

Only about 0.40 of binding-bearing DEV images have every required object detected and binding given presence is defined for 113 of 280 images. Whether this reflects the model or the detector cannot be said before the audit.

13 Discussion

What CFG changes. The clearest result is a dose–response one. Raising w from 2 to 15 makes object presence persistent earlier: under a restart that re-noises by eight steps, a trajectory at $w = 15$ has settled whether its objects appear several steps before a trajectory at $w = 2$ (Section 10). Over the same range, the rate at which the objects appear at all rises and saturates near $w \approx 10$.

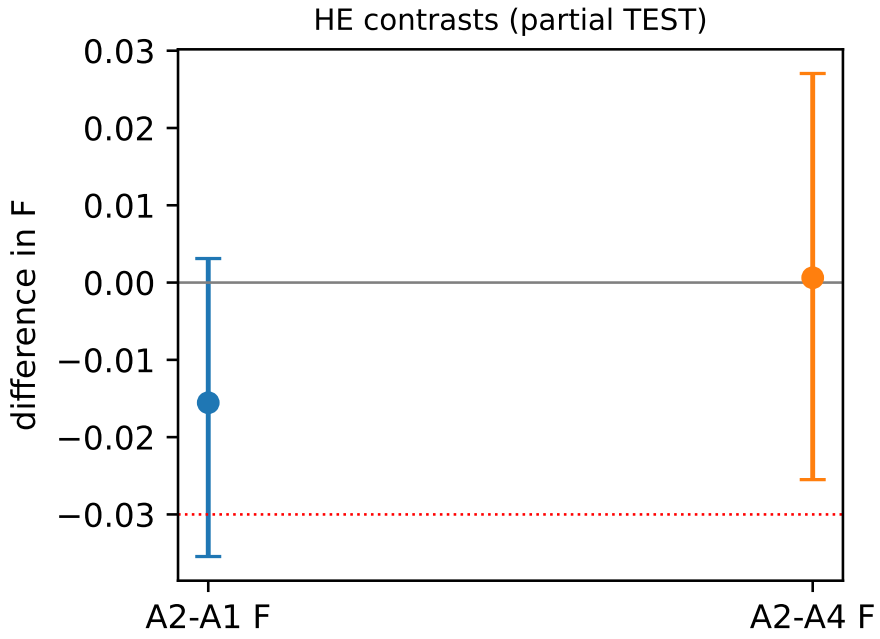


Figure 8: HE contrasts on the composite F: A2−A1 (90% CI, non-inferiority; dotted line is the −0.03 margin) and A2−A4 (95% CI, timing versus the reversed schedule). Partial TEST, exploratory.

Two common worries do not explain the timing effect. Stronger guidance is known to reduce seed diversity, which would raise agreement between any two images. But chance agreement is computed separately at each w, is not higher at $w = 15$ on these scenes, and the uncorrected agreement shows the same ordering. The worry that does survive is measurement noise. Correcting $C(t)$ for a judge that errs independently on 14% of images, the audited rate, removes the difference, so the result is informative only to the extent that the presence judge's errors are content-dependent. Our audit measured how often the judge errs, not how its errors are distributed across seeds. Auditing seed-paired images, which would measure that directly, is the most useful next step for this line of work.

Commitment times are not yet a stable property. The development split suggested an appealing story: colour assignment is fixed at least as early as object presence and numerosity stays open longest. The available test scenes do not reproduce it. Binding becomes persistent after presence, joint binding about eight steps later than on DEV, and the shift appears under both restart probes. Several factors could produce it, and the partial test cannot separate them: a different scene composition, the DEV precision gate that already failed, the unvalidated binding judge, and the left-censoring of the DEV binding curve at the first grid step. We draw one methodological conclusion: a commitment time estimated on about 100 scenes, with intervals of more than ±7 steps, is too imprecise to schedule an intervention on.

Why CAG does not help, and what the controls show. CAG was meant to spend guidance while a constraint is still open and to release it afterwards. Two findings explain why it did not work here. First, its switch steps came from DEV commitment times that did not transfer, so on the test

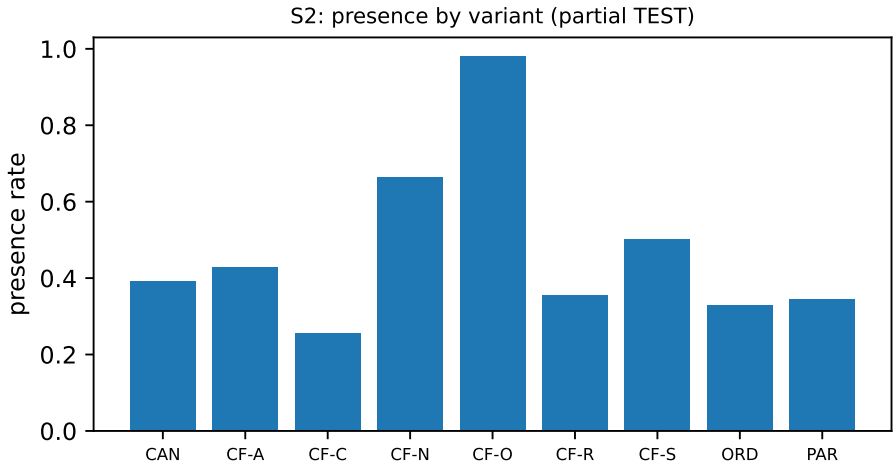


Figure 9: S2: object-presence rate by prompt variant. Partial TEST, exploratory.

scenes it lowers guidance at step 12 while binding is still unsettled. Second, its registered comparison confounds timing with dose. Every CAG arm applies less mean guidance than constant $w = 7.5$, and the dose-matched controls attribute the diversity gain entirely to that lower mean: the constant schedule at the same mean is as diverse, and the reversed schedule more so. At equal mean, front-loading guidance is the *least* diverse of the three orders and no better in adherence. This agrees with interval guidance [16], which reports that guidance at the highest noise levels harms diversity, and with step-selection results [8], where strengthening the conditioning late helps alignment more than doing so early. Our data add a controlled, dose-matched comparison on compositional constraints; they do not show that any timed schedule beats constant guidance of the same mean. Online, prompt-dependent schedules [22] avoid the transfer problem by measuring during sampling, and they are the natural comparison for a revised CAG.

Attention as a cheap proxy. The step at which cross-attention maps stop changing orders constraint types almost opposite to persistence (concordance 0.30, $[0.20, 0.75]$), and within types the scene-level correlations are not distinguishable from zero. On this evidence, attention stabilisation in Stable Diffusion 1.5 is not a substitute for the interventional measurement. This is relevant to methods that read early attention as a forecast of the final image [4, 9]. Their target is image quality or alignment, not the timing of a particular constraint, so our result does not contradict them. It does caution against equating attention that has stopped moving with an outcome that has been decided.

What a reader can take from the null results. Three negative results are informative because the design anticipated them. The full-resample control and the HA precision gate failed on DEV before any test image existed, which is why HA was demoted to exploratory before it could be over-read. HE has a reversed arm and a matched-mean arm, which is what lets us attribute the diversity effect to dose. And the HD stress test shows how precise a judge must be before timing differences of a few steps become interpretable: with independent errors, even 95% accuracy leaves the HD interval touching zero.

Table 17: Robustness of the HD presence contrast $\Delta t_{50} = t_{50}(w{=}15) - t_{50}(w{=}2)$ (steps; scene-bootstrap 95% CI). Judge-error rows correct $C(t)$ for symmetric, non-differential misclassification at rate e (Eq. 6). Partial TEST, exploratory.

Analysis	scenes	Δt_{50}	95% CI
registered estimand ($e = 0$; resampling all 111 scenes)	67	-8.67	[-16.13, -4.87]
resampling only the scenes with $w = 2$ and $w = 15$ branches	67	-8.67	[-16.01, -4.52]
judge error $e = 0.050$	67	-12.91	[-15.49, 0.33]
judge error $e = 0.100$	67	-5.55	[-13.44, 0.00]
judge error $e = 0.144$ (audit)	67	0.00	[-8.59, 0.00]
TEST_CORE shard 0 only	33	-6.64	[-15.97, 5.66]
TEST_CORE shard 1 only	34	-9.66	[-22.27, -5.61]
category: binding	20	-16.44	[-24.00, -4.87]
category: counting	10	11.40	[9.00, 15.00]
category: multi	17	-7.71	[-18.42, 9.71]
category: presence	10	-10.00	[-20.35, 9.33]
category: spatial	10	-0.67	[-12.50, 22.00]

Subgroup rows have few scenes; an interval that includes 0 there is a statement about power, not about the effect.

Table 18: Commitment time t_{50} (steps of 40, $w = 7.5$) on DEV (the values that set $s^\star$) and on the partial TEST split. ≤ 6: left-censored at the first grid step. Only presence has a validated judge. Exploratory.

Type	DEV D8	TEST D8	DEV ER30	TEST ER30
presence	9.3	16.2	9.7	13.3
count	18.9	15.3	15.8	14.0
binding|presence	≤ 6	19.8	≤ 6	21.0
spatial|presence	6.0	≤ 6	≤ 6	≤ 6
size|presence	–	13.6	–	–
joint binding	13.0	21.2	11.5	21.3
scenes	100	111	100	67

DEV sets the frozen switch steps $s^\star = 12$ (presence, binding, spatial prompts) and $s^\star = 21$ (count prompts).

14 Limitations and threats to validity

1. **Incomplete test split.** Three of nine test shards are missing; test results are partial and exploratory, and the registered confirmatory analysis has not been run.

2. **Post-hoc analyses.** Every analysis in Section 10 was chosen after the registered outcomes were known; none is corrected for multiplicity, and subgroup rows (shards, categories) rest on 10–34 scenes.

3. **Judge-error structure.** The HD contrast does not survive an independent-error correction at the audited presence error rate; whether it is real depends on how judge errors are distributed across seeds, which the audit did not measure.

4. **Dose confound in the registered HE contrast.** CAG (mean $w = 4.64$) is registered against constant $w = 7.5$, so HE's diversity component mixes timing with dose; the matched-mean arms A4 and A6 resolve this only in exploratory contrasts.

5. **Transfer of DEV constants.** Commitment times estimated on DEV did not replicate on

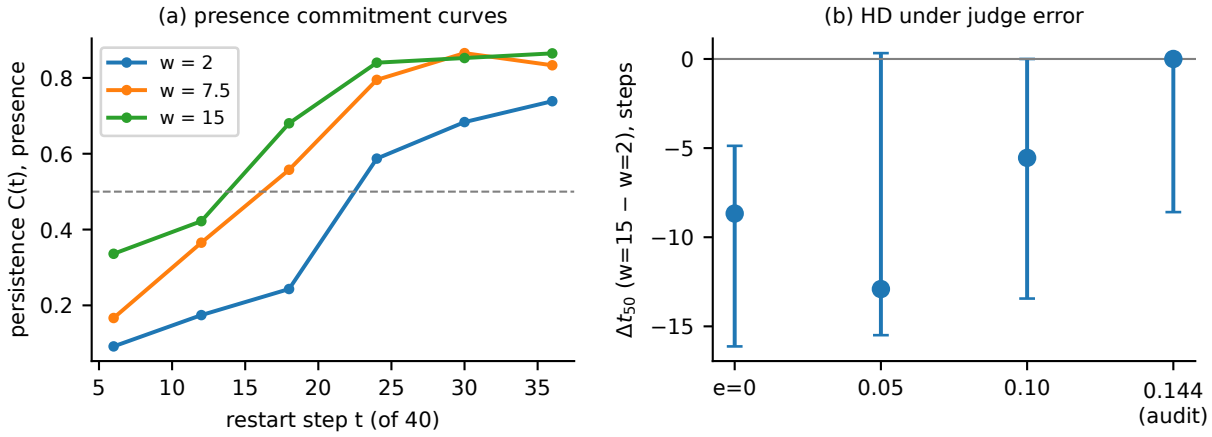


Figure 10: **Partial TEST, exploratory.** (a) Presence persistence $C(t)$ at three guidance scales (restart D8, canonical prompts). (b) The HD contrast $\Delta t_{50} = t_{50}(w{=}15) - t_{50}(w{=}2)$ with 95% scene-bootstrap intervals, without correction ($e = 0$, the registered estimand) and under the independent-error stress test of Eq. (6); at the audited error rate all curves are left-censored and the contrast is 0 by construction.

Table 19: Guidance-schedule contrasts paired by scene (difference, 95% CI); A1 is constant $w = 7.5$. $\bar{w}$: mean applied guidance of the two arms. Presence is the only judged family that met the audit criterion. Partial TEST, exploratory.

Contrast	$\bar{w}$	scenes	ΔF	Δ presence	Δ diversity
A2 (CAG) − A1	4.64 vs 7.50	67	-0.016 [-0.039, 0.006]	-0.026 [-0.052, 0.000]	0.022 [0.015, 0.029]
A3 (global $s^\star$) − A1	4.35 vs 7.50	67	-0.031 [-0.057, -0.008]	-0.045 [-0.082, -0.015]	0.030 [0.022, 0.038]
A4 (reversed) − A1	4.64 vs 7.50	67	-0.021 [-0.062, 0.017]	-0.015 [-0.063, 0.034]	0.040 [0.016, 0.064]
A6 (constant at A2 mean) − A1	4.64 vs 7.50	67	-0.005 [-0.040, 0.030]	-0.015 [-0.063, 0.030]	0.027 [0.009, 0.045]
A2 − A6 (timing at equal dose)	4.64 vs 4.64	134	-0.002 [-0.026, 0.023]	-0.011 [-0.043, 0.021]	-0.013 [-0.026, -0.001]
A2 − A4 (registered timing contrast)	4.64 vs 4.64	134	0.001 [-0.027, 0.027]	-0.015 [-0.049, 0.021]	-0.030 [-0.044, -0.015]

the partial test split, so the frozen switch steps $s^\star$ are mis-timed for binding on the test scenes; CAG is tested here as frozen, not as it would be with accurate switch steps.

6. **HD resampling scope.** The registered HD bootstrap resamples the union of scenes over the three guidance levels (111), of which 67 inform the contrast; restricting to those changes the interval only slightly (Table 17).

7. **Judge validity.** Only presence met the pre-declared accuracy criterion; binding, count, spatial relation and relative size are exploratory. One annotator labelled the audit; no valid double annotation exists, and the pilot-label second review was not done.

8. **Failed development gates.** The HA precision gate and the full-resample control failed; the latter shows an unexplained positive presence offset that would inflate $C(t)$.

9. **Probe dependence.** Restart persistence is defined relative to a re-noising magnitude; under D8 that magnitude varies with t, so absolute t_{50} is not a model property. Several t_{50} values are censored at the first grid step.

10. **Selection.** Conditional outcomes (binding, spatial relation, size given presence) select on a

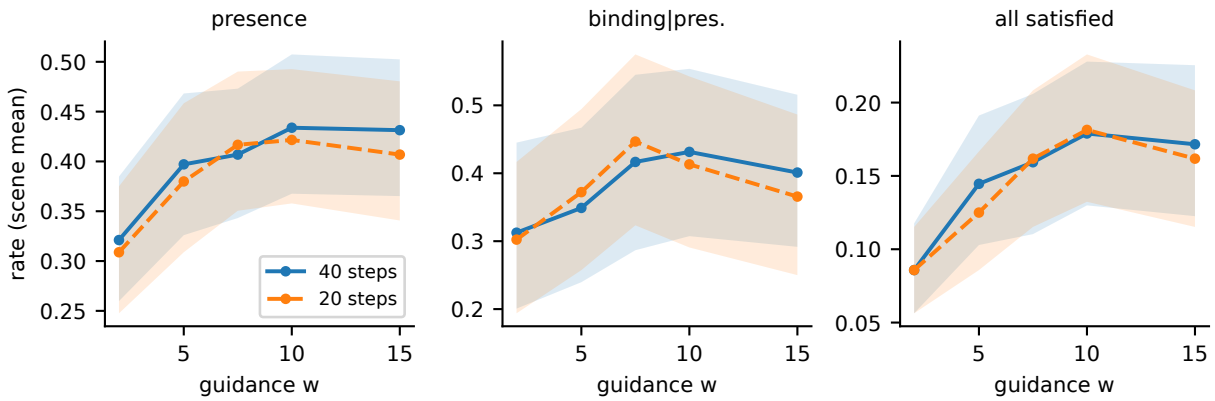


Figure 11: **Partial TEST, exploratory.** Constraint satisfaction on canonical prompts against the guidance scale (scene means over a common set of 102 scenes, 95% scene-bootstrap bands) at 40 and 20 DDIM steps. Binding given presence is shown although its judge missed the audit criterion.

generated variable.

11. **Scope.** One model (Stable Diffusion 1.5), one scheduler and resolution; relative size rests on 24 object pairs; prompts are synthetic and some colour–object pairs are unnatural by design.

12. **Estimator correction.** The chance-agreement term was corrected on DEV before the test split (Section 8); results computed before the correction are archived but not used.

13. **Prior-art review.** Several related works were checked at abstract level only; novelty is not claimed beyond the stated candidate gap.

14. **AI assistance.** Code and documentation were produced with AI assistance; human labels were produced without AI assistance.

15 Reproducibility and provenance

Code is versioned in Git. Every run records its identifier, source commit, pinned model and judge revisions, resolved configuration, software and GPU environment, and SHA-256 checksums of every output; archives are verified file by file before any analysis. The protocol file binds the hashes of 103 code, configuration, benchmark and documentation files, the judge definition, the development-set constants and the three test-stage manifests; test workers and the confirmatory analysis refuse to run unless it verifies. Development constants, audit verdicts and the quality-guard result are stored under `docs/evidence/`; every number in Sections 6–8 is generated from those files by `paper/make_paper_assets.py` and `scripts/paper_dev_tables.py`. Test-split tables and figures are generated by `partial/partial_report.py` (partial) or `scripts/final_report.py` (complete), and the exploratory analyses of Section 10 by `partial/robustness.py`, which reuses the frozen estimator functions on the saved judged rows. The partial-test numbers come from archive `r3-test-partial-20261020T134300Z-40ade4d` (source commit `40ade4d`, protocol verified); an independent re-judging of the same images on different hardware reproduced every judged outcome. Each partial-test number in the text is a generated macro (`tables/partial_robust_numbers.tex`)

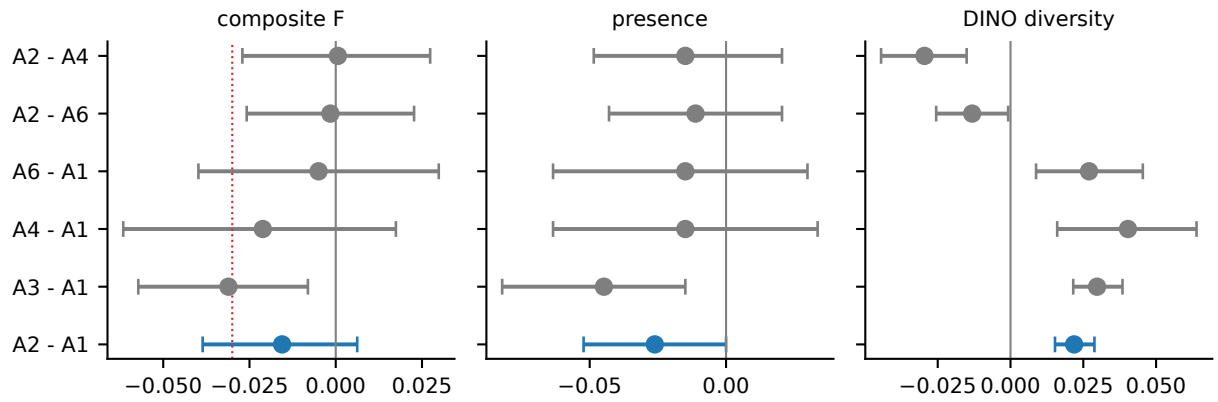


Figure 12: **Partial TEST, exploratory.** Paired scene-level contrasts of the guidance schedules (95% intervals). Rows "Ak − A1" compare against constant $w = 7.5$ (67 scenes with both arms); "A2 − A6" and "A2 − A4" compare schedules with equal mean guidance (134 scenes). Dotted line: the registered non-inferiority margin −0.03 on F.

traceable to `reports/partial/robustness.json` or `partial_results.json`. Decisions and deviations are logged with dates in `docs/R3_DECISION_LOG.md`. Images, latents and human labels are not distributed.

16 Conclusion

We asked when prompt constraints become persistent during diffusion sampling, what classifier-free guidance changes about that timing, whether cross-attention reveals it, and whether a guidance schedule timed by the measurement is useful. We answered with an interventional, chance-corrected estimator under a signed pre-registration, and reported every gate as it fell.

On the six available test shards, all exploratory: stronger guidance makes object presence persistent earlier, the only registered rule that passed, but the result holds only if the presence judge's errors are not independent across seeds. Commitment times measured on the development split did not transfer to the test scenes; attention stabilisation did not track persistence; and Commitment-Aligned Guidance, scheduled from those development times, was neither non-inferior to constant guidance nor better than its reverse. Its diversity gain is a dose effect, which matched-mean controls exposed.

The broader lesson concerns precision. Timing differences of a few steps are only interpretable with a judge that is substantially more accurate than current detector-based judges for anything beyond presence, and with commitment estimates far tighter than about 100 development scenes provide. Two things remain: the registered confirmatory analysis, which needs the three missing test shards, and a seed-paired judge audit, which would decide whether the guidance-timing effect is real.

References

[1] Yogesh Balaji, Seungjun Nah, Xun Huang, et al. eDiff-I: Text-to-image diffusion models with an ensemble of expert denoisers, 2022. arXiv:2211.01324; content claim UNVERIFIED this session.

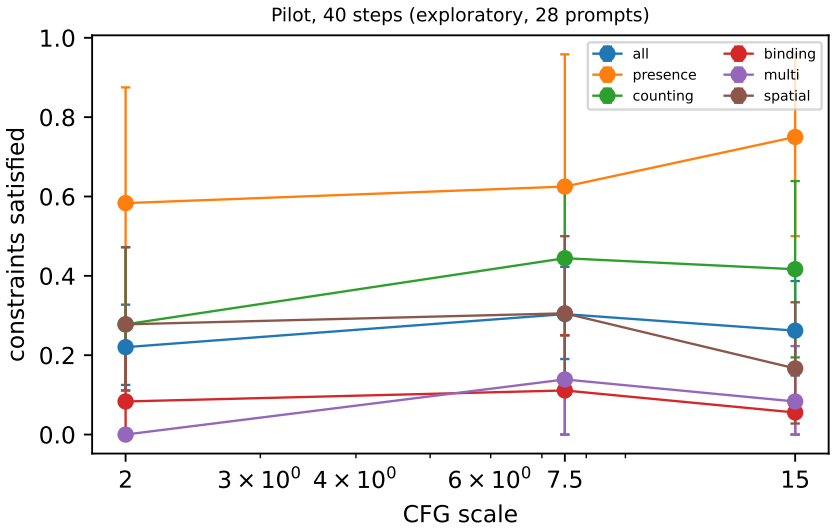


Figure 13: Pilot constraint satisfaction against CFG (40 steps, exploratory).

[2] Yuanhao Ban, Ruochen Wang, Tianyi Zhou, Minhao Cheng, Boqing Gong, and Cho-Jui Hsieh. Understanding the impact of negative prompts: When and how do they take effect?, 2024. arXiv:2406.02965.

[3] Hila Chefer, Yuval Alaluf, Yael Vinker, Lior Wolf, and Daniel Cohen-Or. Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. *ACM Transactions on Graphics*, 42(4), 2023. arXiv:2301.13826.

[4] Benlei Cui, Bukun Huang, Zhizeng Ye, Xuemei Dong, Tuo Chen, Hui Xue, Dingkang Yang, Longtao Huang, Jingqun Tang, and Haiwen Hong. Diffusion probe: Generated image result prediction using CNN probes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 35926–35935, 2026. arXiv:2602.23783.

[5] Dave Epstein, Allan Jabri, Ben Poole, Alexei A. Efros, and Aleksander Holynski. Diffusion self-guidance for controllable image generation. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.00986; content not re-verified this session.

[6] Yuhan Fu and Gong Cheng. Enhancing semantic mapping in text-to-image diffusion via gather-and-bind. *Computers & Graphics*, 125:104118, 2024. check author list against the journal page.

[7] Dhruba Ghosh, Hannaneh Hajishirzi, and Ludwig Schmidt. GenEval: An object-focused framework for evaluating text-to-image alignment. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks*, 2023. arXiv:2310.11513.

[8] Paul Grimal, Hervé Le Borgne, and Olivier Ferret. Text-to-image alignment in denoising-based models through step selection, 2025. arXiv:2504.17525.

[9] Xiefan Guo, Jinlin Liu, Miaomiao Cui, Jiankai Li, Hongyu Yang, and Di Huang. InitNO: Boosting text-to-image diffusion models via initial noise optimization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2404.04650.

Table 20: Pilot (28 prompts, detector-based labels, exploratory): fraction of constraints satisfied, by category, CFG and inference steps; 95% prompt-cluster bootstrap intervals.

Category	CFG	20 steps	40 steps	Prompts
all	2	0.24 [0.137, 0.351]	0.22 [0.125, 0.327]	28
all	7.5	0.30 [0.190, 0.417]	0.30 [0.190, 0.423]	28
all	15	0.32 [0.202, 0.440]	0.26 [0.149, 0.387]	28
presence	2	0.58 [0.292, 0.875]	0.58 [0.292, 0.875]	4
presence	7.5	0.71 [0.417, 0.958]	0.62 [0.292, 0.958]	4
presence	15	0.67 [0.292, 1.000]	0.75 [0.500, 0.958]	4
counting	2	0.36 [0.167, 0.583]	0.28 [0.111, 0.472]	6
counting	7.5	0.39 [0.194, 0.611]	0.44 [0.250, 0.639]	6
counting	15	0.44 [0.250, 0.639]	0.42 [0.194, 0.639]	6
binding	2	0.06 [0.000, 0.194]	0.08 [0.000, 0.222]	6
binding	7.5	0.11 [0.000, 0.278]	0.11 [0.000, 0.250]	6
binding	15	0.17 [0.028, 0.333]	0.06 [0.000, 0.167]	6
multi	2	0.00 [0.000, 0.000]	0.00 [0.000, 0.000]	6
multi	7.5	0.11 [0.000, 0.333]	0.14 [0.000, 0.306]	6
multi	15	0.08 [0.000, 0.250]	0.08 [0.000, 0.223]	6
spatial	2	0.31 [0.111, 0.500]	0.28 [0.111, 0.472]	6
spatial	7.5	0.31 [0.139, 0.500]	0.31 [0.111, 0.500]	6
spatial	15	0.33 [0.167, 0.528]	0.17 [0.028, 0.333]	6

Source: `outputs/main/results/adherence_by_cell.csv`. Intervals resample prompts; the judge is unaudited on the DEV set.

[10] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control. In *International Conference on Learning Representations*, 2023. arXiv:2208.01626.

[11] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[12] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2006.11239.

[13] Yushi Hu, Benlin Liu, Jungo Kasai, Yizhong Wang, Mari Ostendorf, Ranjay Krishna, and Noah A. Smith. TIFA: Accurate and interpretable text-to-image faithfulness evaluation with question answering, 2023. arXiv:2303.11897.

[14] Kaiyi Huang, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2I-CompBench: A comprehensive benchmark for open-world compositional text-to-image generation. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks*, 2023. arXiv:2307.06350.

[15] Amita Kamath, Kai-Wei Chang, Ranjay Krishna, Luke Zettlemoyer, Yushi Hu, and Marjan Ghazvininejad. GenEval 2: Addressing benchmark drift in text-to-image evaluation, 2025. arXiv:2512.16853.

[16] Tuomas Kynkäänniemi, Miika Aittala, Tero Karras, Samuli Laine, Timo Aila, and Jaakko Lehtinen. Applying guidance in a limited interval improves sample and distribution quality in diffusion models. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2404.07724.

[17] Marvin Li and Sitan Chen. Critical windows: non-asymptotic theory for feature emergence in diffusion models. In *International Conference on Machine Learning*, 2024. check metadata.

Table 21: Pilot CFG contrasts (paired by prompt and seed; Holm-adjusted p from Wilcoxon tests), exploratory.

Category	CFG a vs b	Diff.	95% CI	Prompts	p_{Holm}
presence	2 vs 7.5	-0.083	[-0.438, 0.229]	4	1.000
presence	2 vs 15	-0.125	[-0.500, 0.250]	4	1.000
presence	7.5 vs 15	-0.042	[-0.208, 0.104]	4	1.000
binding	2 vs 7.5	-0.042	[-0.181, 0.097]	6	1.000
binding	2 vs 15	-0.042	[-0.153, 0.083]	6	1.000
binding	7.5 vs 15	+0.000	[-0.111, 0.125]	6	1.000
counting	2 vs 7.5	-0.097	[-0.292, 0.083]	6	1.000
counting	2 vs 15	-0.111	[-0.333, 0.083]	6	1.000
counting	7.5 vs 15	-0.014	[-0.097, 0.056]	6	1.000
spatial	2 vs 7.5	-0.014	[-0.167, 0.153]	6	1.000
spatial	2 vs 15	+0.042	[-0.069, 0.125]	6	0.750
spatial	7.5 vs 15	+0.056	[-0.069, 0.181]	6	1.000
multi	2 vs 7.5	-0.125	[-0.292, 0.000]	6	0.750
multi	2 vs 15	-0.083	[-0.222, 0.000]	6	1.000
multi	7.5 vs 15	+0.042	[-0.083, 0.208]	6	1.000
all	2 vs 7.5	-0.071	[-0.155, 0.009]	28	0.174
all	2 vs 15	-0.060	[-0.146, 0.018]	28	0.214
all	7.5 vs 15	+0.012	[-0.045, 0.071]	28	0.873

[18] Yumeng Li, Margret Keuper, Dan Zhang, and Anna Khoreva. Divide & bind your attention for improved generative semantic nursing. In *British Machine Vision Conference*, 2023. arXiv:2307.10864.

[19] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. SDEdit: Guided image synthesis and editing with stochastic differential equations. In *International Conference on Learning Representations*, 2022. check.

[20] Matthias Minderer, Alexey Gritsenko, Austin Stone, Maxim Neumann, Dirk Weissenborn, Alexey Dosovitskiy, Aravindh Mahendran, Anurag Arnab, Mostafa Dehghani, Zhuoran Shen, Xiao Wang, Xiaohua Zhai, Thomas Kipf, and Neil Houlsby. Simple open-vocabulary object detection with vision transformers. In *European Conference on Computer Vision (ECCV)*, 2022. OWL-ViT; arXiv:2205.06230.

[21] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. diversity embedder (facebook/dinov2-small).

[22] Pinelopi Papalampidi, Olivia Wiles, Ira Ktena, Aleksandar Shtedritski, Emanuele Bugliarello, Ivana Kajic, Isabela Albuquerque, and Aida Nematzadeh. Dynamic classifier-free diffusion guidance via online feedback, 2025. arXiv:2509.16131.

[23] Or Patashnik, Daniel Garibi, Idan Azuri, Hadar Averbuch-Elor, and Daniel Cohen-Or. Localizing object-level shape variations with text-to-image diffusion models. In *IEEE/CVF International Conference on Computer Vision*, 2023. arXiv:2303.11306; prompt-mixing operation; full text to re-read.

[24] Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning transferable visual models

Table 22: Pilot global attention re-weighting (scale 2 and 4 against the scale-1 control path), primary outcome, exploratory.

Category	Arm	Diff.	95% CI	Prompts	p_{Holm}
presence	rw2 vs control	+0.167	[0.000, 0.375]	4	0.500
presence	rw4 vs control	+0.125	[-0.042, 0.333]	4	0.500
binding	rw2 vs control	+0.028	[-0.111, 0.167]	6	1.000
binding	rw4 vs control	+0.028	[-0.139, 0.222]	6	1.000
counting	rw2 vs control	+0.028	[0.000, 0.111]	6	1.000
counting	rw4 vs control	+0.028	[-0.083, 0.139]	6	1.000
spatial	rw2 vs control	-0.028	[-0.194, 0.139]	6	1.000
spatial	rw4 vs control	+0.056	[-0.139, 0.250]	6	1.000
multi	rw2 vs control	-0.028	[-0.194, 0.111]	6	1.000
multi	rw4 vs control	-0.083	[-0.222, 0.000]	6	1.000
all	rw2 vs control	+0.024	[-0.042, 0.089]	28	0.757
all	rw4 vs control	+0.024	[-0.054, 0.095]	28	0.757

Control images were pixel-identical to the unmodified pipeline (owner-documented 168/168).

Table 23: Pilot nested prediction of the primary outcome (40-step images, prompt-grouped cross-validation, 28 prompts); exploratory.

Contrast	ΔAUC	95% prompt-bootstrap CI
attn_minus_design	+0.067	[0.028, 0.122]
noise_minus_design	+0.002	[-0.028, 0.027]
attn_minus_noise	+0.050	[0.017, 0.097]

Base rate 0.236. Only six distinct noise vectors exist in the pilot, so noise features equal seed identity (noise vs design: -0.007).

from natural language supervision. In *International Conference on Machine Learning*, 2021. arXiv:2103.00020.

[25] Royi Rassin, Eran Hirsch, Daniel Glickman, Shauli Ravfogel, Yoav Goldberg, and Gal Chechik. Linguistic binding in diffusion models: Enhancing attribute correspondence through attention map alignment. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.08877; check.

[26] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. arXiv:2112.10752.

[27] Seyedmorteza Sadat, Jakob Buhmann, Derek Bradley, Otmar Hilliges, and Romann M. Weber. CADS: Unleashing the diversity of diffusion models through condition-annealed sampling. In *International Conference on Learning Representations*, 2024. arXiv:2310.17347.

[28] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021. arXiv:2010.02502.

[29] Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting stable diffusion using cross attention. In *Proceedings of the Association for Computational Linguistics*, 2023. arXiv:2210.04885.

Table 24: Measured throughput on Tesla T4 (seconds per 40-step image, first image excluded) from two probe sessions; the real R2a run gave a median of 6.21 and 6.26 s per worker.

Session	Plain	With latents	Two workers, wall (s)
R1 session 10:10 UTC	6.16	6.25	53.9
R0 session 15:02 UTC	7.76	7.59	60.5

Table 25: R2a DEV core, first-pass detector (J1) outcomes on binding-bearing canonical prompts. **Unaudited judge; exploratory.**

Quantity	Value	n
All required objects detected	0.404	280
Binding given presence	0.487	113
Joint (present and bound)	0.196	280

70 scenes × 4 seeds. Human error rates of the judge are pending (§6).

[30] Yilun Xu, Mingyang Deng, Xiang Cheng, Yonglong Tian, Ziming Liu, and Tommi Jaakkola. Restart sampling for improving generative processes. In *Advances in Neural Information Processing Systems*, 2023. check.

[31] Yuxin Zhang, Weiming Dong, Fan Tang, et al. ProSpect: Prompt spectrum for attribute-aware personalization of diffusion models. *ACM Transactions on Graphics (SIGGRAPH Asia)*, 2023. arXiv:2305.16225; check author list.